

Questions on RESfM

To: Fadi Khatib (RESfM, first author)

Subject: Follow-up

Dear Fadi,

Thank you again for the earlier discussion and for the metric-priority guidance. I am working on an unsupervised / self-supervised approach to outlier handling for equivariant SfM, building directly on RESfM, and I have reached a point where a few specifics about your setup would materially change what I run next. I have ordered them by how much each answer would help; the first four are the ones I would most value.

1. Your preprocessed point tracks for the cross-dataset experiments (1DSfM, Strecha, BlendedMVS).

I reconstructed the point tracks for the published cross-dataset experiments in Table 2, highlighting the results fitted to your setup. The like-for-like comparison reproduces the differences in the match. If your construction of the tracks, my

2. Your trained ESfM and GASfM weights.

Section 4.2 of the paper, including the ESfM and GASfM weights. Loading the weights is unexplained.

3. GLOMAP (and Theia / COLMAP) versions and settings, and how the precomputed tracks were passed in.

Section 4.3 of the paper, from feature extraction to baseline I re-ran (3.98) while

4. The metric priority you gave me - translation, then rotation, then registered cameras - is it lexicographic or weighted?

That is, do you give rotation a higher claim, since

If you have a little more time, four further questions would help:

5. Running your released checkpoint on a few test scenes, what gap to the published numbers would you consider normal rather than a sign that my evaluation path is wrong?

6. What was the outlier-percentage distribution across the 27 training and 4 validation

scenes?

The paper
about whet

7. Did you ever regenerate labels on a target domain such as 1DSfM and retrain the outlier head there?

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8. How much of the outlier robustness comes from the learned head versus the robust BA?

Table 11 s
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Thank you very much - even partial answers to 1, 4, 7 and 8 would save me significant compute and sharpen the claim.

Best regards,
Ortal Dayan